THE COPPERBELT UNIVERSITY

SCHOOL OF MATHEMATICS AND NATURAL SCIENCES

SEPTEMBER, 2012 – EARLY SESSIONAL EXAMINATIONS

CS334 – DATABASE TECHNOLOGY

TIME ALLOWED: THREE HOURS

INSTRUCTIONS :

- **Maximum Marks Available – 100**
- **This Paper has SIX(6) Questions**
- **Question One is Compulsory and you are required to Answer Any other FOUR (4) from the remaining FIVE(5) Questions**
- **All Questions Carry Equal Marks (20 marks)**

Q 2 a) Consider the LIBRARY relational database schema shown in the given Figure below and write down relational algebra expressions for the following queries. **[10 marks]**

i) List the details of borrowers who borrowed books from both the “CBU library” and the “Kitwe City council Library”

ii) List the details of Branch libraries that have copies of a book titled “Management Information Systems” but do not have a book with the title “Introduction to databases”.

iii) List the details of authors that have supplied their books to either the “CBU Library” or the 'Kitwe City council Library.'

iv) For each book that is loaned out from the 'CBU Library' branch and whos DueDate is today, retrieve the book title, the borrower's details.

b) Using the same LIBRARY relational database schema shown in the given Figure below, write down SQL statements for the following queries. **[10 marks]**

i) Find the details of all the publishers and the number of books they have published.

ii) Find the number of books borrowed by each borrower and output the details of borrowers that have borrowed more than 4 books from the “CBU Library”.

iii) Find the details for each author that has contributed more than 2 different books at the CBU library, the output should be in ascending order using the author’s names.

iv) Find the number of copies of the book titled “*Internet Technologies”* owned by each library branch.

Q 2 a) Define the following database terminologies **[6 marks]**

i) Redundancy duplication ii) Entity iii) System catalog

iv) Conceptual Schema v). Data consistency vi) Data integrity

b) Expand on what is meant by the following DB system disadvantages. **[8 marks]**

i). Additional hardware costs ii). Cost of conversion iii) Performance
iv) Higher impact of a failure

c) Briefly distinguish between file based systems and database systems. **[6 marks]**

Q 3 The following statement describes a version of a University database. Draw an EER diagram that shows the entity types, attributes, nature of relationships, and specializations for this application. State any assumptions you make. **[20 marks]**

University Database Application

For each person, the database maintains information on the person's Name, NRC no, address, sex, and birth date. The person can either be a Faculty member or a student. A faculty member has a rank, office phone number and Salary. All faculty members are related to the academic department(s) with which they are affiliated. A faculty member can be associated with one or several departments. A Student on the other hand has year of study. Each student is also related to his or her department. A student is registered to a programme and each programme has courses. A programme is identified by programme code and programme name. A course on the other hand is identified by course code and course name. Upon completion a student is given a Transcript. Each transcript instance includes the courses taken by a student and the grade the student received in the course section. A student can either be a graduate or undergraduate student. For each graduate student, we keep a list of previous degrees earned. We also relate the graduate student to a faculty advisor. An academic department is identified by name, telephone and office number and is related to the faculty member who is its Head and to the School to which it belongs. Each School has a name, office number, and the name of its dean.

Q 4 Map the EER model obtained in Q 3 above into relational schemas. Explain your mapping process **[20 marks].**

Q 5 a) Define Boyce-Codd normal form. How does it differ from 3NF? Why is it considered a stronger form of 3NF? **[5 marks]**

b) Consider the universal relation G with the following different set of functional dependencies.

$$G = \{\{A,B\} \rightarrow \{C\}, \{B,D\} \rightarrow \{E,F\}, \{A,D\} \rightarrow \{G,H\}, \{A\} \rightarrow \{I\}, \{H\} \rightarrow \{J\}\}$$

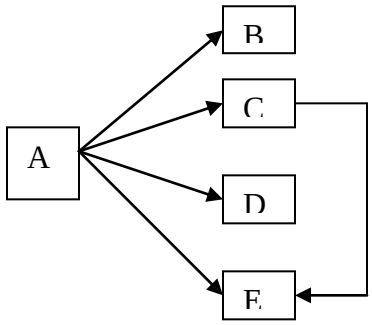
Determine the primary key for G and hence or otherwise decompose G into 2NF and then 3NF relations. **[10 marks]**

c) Consider a relation R(A, B, C, D, E) with the following dependencies: $AB \rightarrow C,\ CD \rightarrow E,\ DE \rightarrow B$

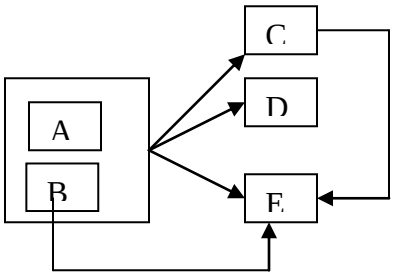
Is AB a candidate key of this relation? If not, is ABD? Explain your answer.

[5 marks]

i)



ii)



b) State atleast four (4) benefits of normalizing database tables**. [4 marks]**

c) Consider the relation R = {A, B, C, D, E, F, G, H, I,J} and the set of functional dependencies F ={{A, B} ⟶ {C}, {A} ⟶ {D, E}, {B} ⟶ {F}, {F} ⟶ {G, H},{D} ⟶ {I, J }},

Determine the key for R and hence or otherwise decompose R into 2NF and then 3NF relations. **[6 marks]**

d) When is a table said to be in Boyce Codd Normal Form (BCNF) and hence or otherwise normalize the table below to BCNF. **[6 marks]**

Student	Subject	Lecturer
1122679	Web Design	Mr Shumba
1122679	Computer Networks	Mr Lengwe
1023456	Web Design	Mr Shumba
1045678	Internet Tech	Dr Libati
1167859	Web Design	Mr Nkandu

Q 4 a) List the operations of relational algebra and the purpose of each. **[8 marks]**

b) Discuss the following forms of the Join operation in Relational Algebra**.**
[4 marks]

i) Theta Join ii) Equijoin iii) Semi Join
iv) Natural Join

b) Given the database schema, write equivalent relational algebra statements for the queries below**. [8 marks]**

Employee (Fname,Lname, SSN, Bdate, Address, Sex, Salary, SuperSSN, DNO)
Department (Dname, Dnumber, MngrSSN)
Dept-Location (Dnumber, Dlocation)
Project (Pname, Pnumber, Plocation, Dnum)
Works-On (ESSN, Pno, Hours)
Dependent (ESSN, Dependent_Name, Sex, Bdate, Relationship).

i) List the names of all employees who have a dependent with the same first name as themselves.
ii) Find the names of all employees who are supervised by 'Mr Zulu'.
iii) List the last names of all department managers who have no dependents.
iv) For each project, list the project name and the total hours per week (by all employees) spent on that project.

Q 5 a) State briefly how SQL can allow implementation of the entity integrity and referential integrity constraints. **[4 marks]**

b) Briefly describe the six clauses in the syntax of an SQL query, and show which of the six clauses are required and which are optional. **[4 marks]**

c) Write appropriate SQL DDL statements for declaring the relational database schema in Q 4 b) above. Specify appropriate keys and referential triggered actions. **[12 marks]**

Q 6 a) Write relational algebra statements for the queries in Q 4 b) above. **[8 marks]**

b) To ensure that a transaction preserves the consistency of the database, a transaction must be atomic, execute to completion and not execute at all. Briefly discuss the above statement. **[4 marks].**

c) Briefly discuss atleast four (4) facilities provided by DBMS to recover from failure. **[4 marks]**

d) State atleast four (4) specific functionalities of DDBMS. **[4 marks].**

……**THE END**…….